ARCHITECTURE & OPERATIONAL GUIDANCE

Azure Monitoring and Logging Strategy

OVERVIEW

Our Azure applications and services require a comprehensive monitoring strategy designed to maintain high availability, accelerate incident resolution, and ensure strict compliance. The goal is not to choose a single tool and discard others, but rather to leverage each platform for its optimal purpose.

> **Key Objective:** Establish clear operational boundaries and integration flows between native Azure telemetry tools and our enterprise Splunk deployment to achieve optimal performance and cost efficiency.

Current Ecosystem Capabilities:

- **Azure Monitor:** Native platform metrics and resource health indicators.
- **Application Insights:** Deep APM, distributed tracing, and code-level diagnostics.
- **Log Analytics Workspace (LAW):** Centralized log query data store powered by KQL.
- **Splunk (Enterprise-Managed):** Multi-cloud security, compliance, and long-term retention platform.

SIMPLE VIEW OF THE TOOLS

Azure Monitor

The first place to check if an Azure resource is healthy or unhealthy. Monitors infrastructure metrics, CPU, memory, response times, and resource availability.

Best For: Infrastructure monitoring, platform alerts, native metrics, and resource health.

Application Insights

Focused entirely on application performance and troubleshooting. Answers why apps are slow, which APIs or database queries failed, and traces dependencies.

Best For: APM, root cause analysis, performance diagnostics, and distributed end-to-end tracing.

Log Analytics Workspace (LAW)

Azure's central log repository. Stores telemetry from Application Insights and diagnostic logs from Azure services. Workspace-based App Insights uses LAW as its data store.

Best For: Log searching, KQL queries, technical troubleshooting, log alerts, and Azure Workbooks.

Splunk (Enterprise)

Enterprise-wide logging platform collecting telemetry across diverse environments (Azure, AWS, VMware, Linux, Windows, Firewalls, and Network devices).

Best For: Enterprise visibility, SOC/security monitoring, compliance reporting, and long-term retention.

MONITORING DATA FLOW

1. Application Monitoring Flow

Application telemetry flows seamlessly from code-level instrumentation into Application Insights, which persists the structured log data inside Log Analytics Workspace for analytical querying and dashboarding.

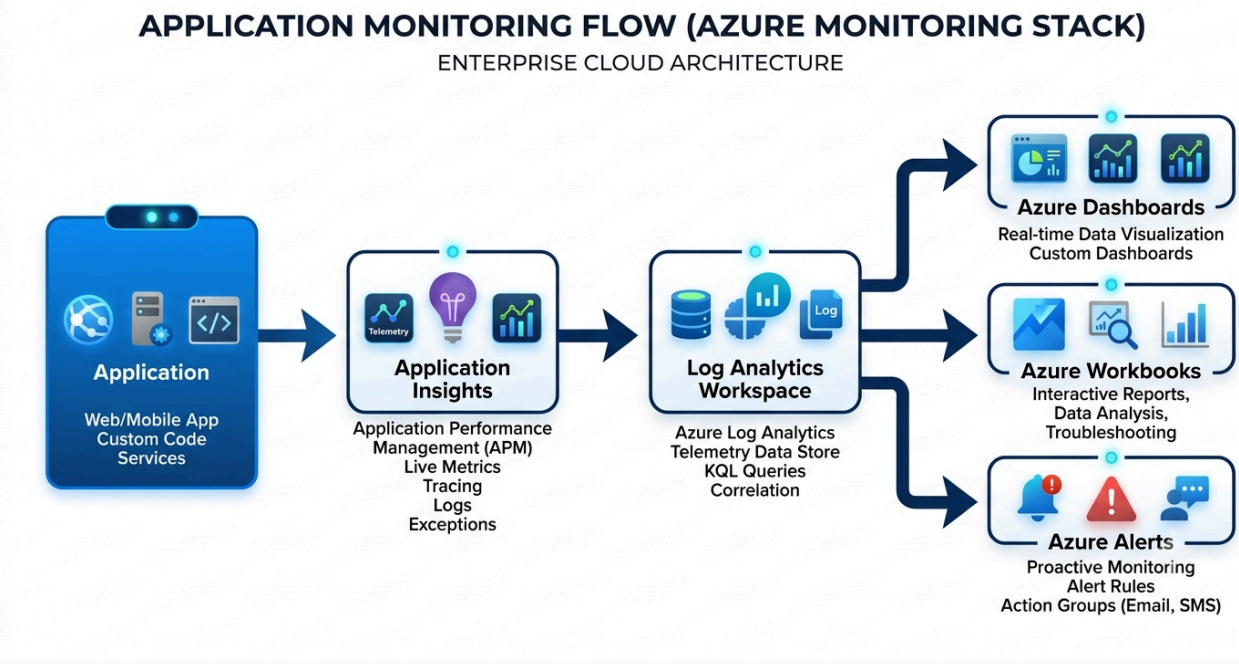
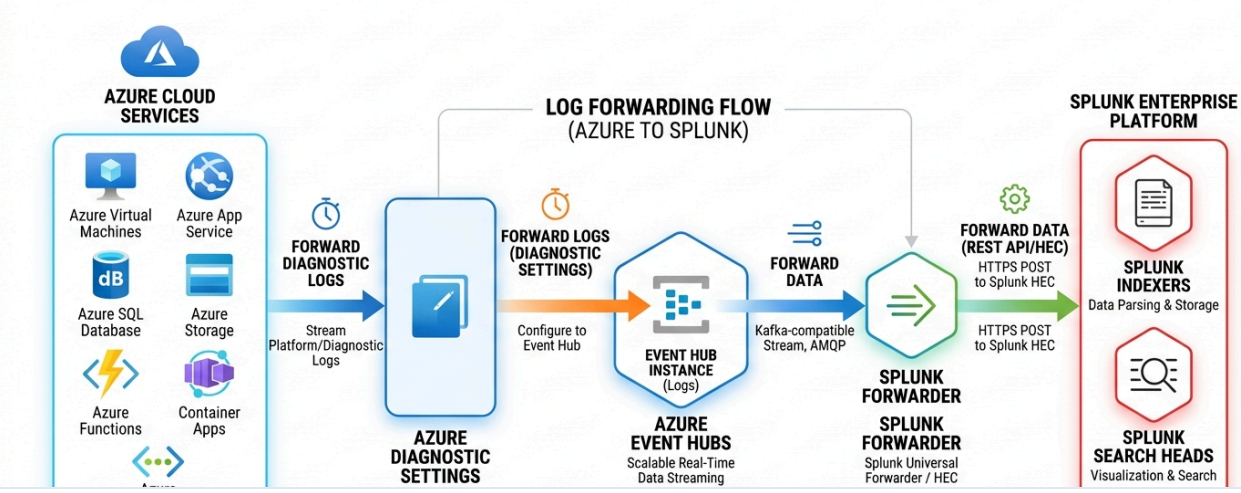


Figure 1: End-to-end telemetry pipeline from Azure application components into Log Analytics, Dashboards, Workbooks, and Alerts.

> **Example Troubleshooting Flow (User Login Failure):**
> User Login Failed → App Insights captures exception → Stored in LAW → Engineer executes KQL query → Root cause identified in seconds.

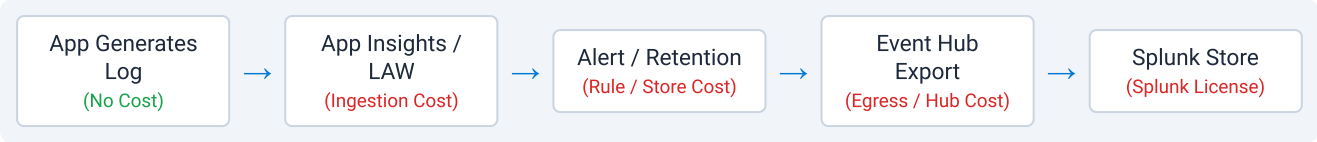
2. Infrastructure Monitoring & Splunk Integration Flow

Azure resource diagnostic settings stream log data simultaneously to native Log Analytics Workspaces for immediate ops support and to Azure Event Hub for downstream ingestion into Splunk.



WHERE DOES AZURE COST START?

Standard platform metrics and Activity Logs are collected at no additional cost by Microsoft. Primary Azure Monitor costs stem from log ingestion, extended retention, log-based alerts, and data export streams.



Included at No Additional Cost

- ✅ Azure Activity Logs
- ✅ Standard Azure Metrics
- ✅ Basic metric viewing & charts
- ✅ Platform metrics collection
- ✅ Dashboard & Workbook creation

Billable Components

- **LAW Data Ingestion:** Per-GB charged upon intake.
- **App Insights:** Charged based on LAW ingestion volume.
- **Alert Rules:** Metric & log query execution frequency.
- **Event Hub:** Throughput units & ingress/egress for Splunk.

SAMPLE MONTHLY COST CALCULATION

Note: Industry benchmark reference estimates standard analytics log ingestion at ~$2.30/GB. Regional prices vary.

Environment Size	Daily Volume	Monthly Volume	Estimated Monthly Ingestion Cost
Small Environment	10 GB / day	300 GB / month	**$690 / month**
Medium Environment	100 GB / day	3,000 GB / month	**$6,900 / month**
Large Environment	500 GB / day	15,000 GB / month	**$34,500 / month**

COMPARATIVE ADVANTAGE & POSITIONING

Why Splunk Is Better

- **Enterprise Visibility:** Single pane for Azure, AWS, On-prem, and Network logs.
- **Security & SOC:** Advanced threat hunting, SIEM, and SOC investigation workflow.
- **Compliance:** Multi-year regulatory data retention (1–3+ years).

Why LAW Is Better

- **Azure-Native Diagnostics:** Deep integration with App Service, SQL, AKS, and Key Vault.
- **Rapid APM Troubleshooting:** Immediate end-to-end trace correlation and dependency maps.
- **Operator Speed:** Fast KQL querying for day-to-day Azure engineering support.

TARGET ROUTING & RETENTION STRATEGY

Category	Log / Telemetry Type	Primary Target	Recommended Retention
Applications	Exceptions, Requests, Dependencies, Traces	LAW (App Insights)	30 – 90 Days
Azure Resources	App Service, SQL, Key Vault, Storage, AKS	LAW	30 – 90 Days
Security Logs	Firewall logs, Security events, Authentication	Splunk	1 – 3 Years
Compliance Logs	Audit trails, Governance, Multi-cloud activity	Splunk	1 – 3 Years

STRATEGY & OWNERSHIP MATRIX

Azure Team Ownership

- **Platform:** Azure Monitor, App Insights, LAW
- **Artifacts:** Workbooks, Azure Dashboards, Metric Alerts
- **Governance:** Diagnostic Settings, Azure Policy enforcement
- **Focus:** Operational health & fast troubleshooting

Splunk Team Ownership

- **Platform:** Splunk Enterprise & Forwarders
- **Artifacts:** Enterprise SIEM dashboards & SOC alerts
- **Governance:** Security retention rules & ingestion caps
- **Focus:** Cross-cloud security, compliance & audit

FINAL STRATEGIC DECISION

Coexistence Model Architecture

Splunk is not a replacement for Log Analytics Workspace, and LAW is not a replacement for Splunk.

Log Analytics Workspace remains the primary operational monitoring and troubleshooting environment for Azure engineering teams, ensuring the fastest mean-time-to-resolution (MTTR). Splunk serves as the centralized enterprise logging, security, and compliance platform. This dual-purpose strategy maximizes operational speed while fully honoring enterprise security standards.